

Exact circle shape optimization for geodesic observation and for resonant wave times

Candidate partial answer to the shape-optimization problems of arXiv:1607.01535

11 August 2026

Status: candidate partial result, likely valid. On the circle, the finite-time geometric functional g_2^T has supremum L for every observation time, with a complete attainment classification by rational rotations. At the resonant times $T \in 2\pi\mathbb{N}$, the deterministic wave constant satisfies $C_T = (T/2)g_1$ exactly; its supremum is $TL/2$ and it is attained precisely at volume fraction $L = 1/2$ by antipodal selectors.

Source problem

Humbert, Privat, and Trélat ask on PDF page 22 of *Observability properties of the homogeneous wave equation on a closed manifold* [1]:

A challenging problem is to maximize $\omega \mapsto C_T(\omega)$ over all measurable subsets of prescribed volume fraction L . Maximizing the geometric functional $\omega \mapsto g_2(\omega)$ is also an interesting open problem and would be a step toward the deterministic problem.

The source passage is reproduced below.

developed in this paper, combined with microlocal issues, may serve to address this problem.

Shape optimization. A challenging problem is to maximize the functional $\omega \mapsto C_T(\omega)$ over the set of all possible measurable subsets of Ω of measure $|\omega| = L|\Omega|$ for some fixed $L \in (0, 1)$. In [24, 26], the maximization of the *randomized* observability constant has been considered, that is, the functional $\omega \mapsto g_1(\omega)$. Maximizing the functional $\omega \mapsto g_2(\omega)$ is an interesting open problem which, thanks to Corollary 4, would be a step towards the maximization of the *deterministic* observability constant.

References

- [1] C. Bardos, G. Lebeau, J. Rauch, *Sharp sufficient conditions for the observation, control, and stabilization of waves from the boundary*, SIAM J. Control Optim. **30** (1992), no. 5, 1024–1065.
- [2] N. Burq, P. Gérard, *Condition nécessaire et suffisante pour la contrôlabilité exacte des ondes (French)* [A

We solve the complete circle subcase for the long-time functional g_2 , and more strongly for its finite-time version g_2^T . We also solve the deterministic problem at every common spectral period. The arbitrary-time deterministic problem and higher-dimensional manifolds remain open.

Definitions

Write $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, with arc-length measure, and

$$\mathcal{U}_L = \{\omega \subset \mathbb{T} : \omega \text{ measurable and } |\omega| = 2\pi L\}, \quad 0 < L < 1.$$

Every unit-speed geodesic is $t \mapsto x \pm t$. Hence the source's geometric quantity is

$$g_2^T(\omega) = \inf_{x \in \mathbb{T}} \frac{1}{T} \int_0^T \chi_\omega(x+t) dt. \quad (1)$$

The sign of the ray is immaterial after changing x .

For the wave equation on the zero-mean sector used in the source's circle example, let $C_T(\omega)$ be its deterministic observability constant and let

$$g_1(\omega) = \inf_{j \geq 1} \inf_{0 \neq \phi \in \text{span}\{\cos(jx), \sin(jx)\}} \frac{\int_\omega |\phi|^2}{\int_{\mathbb{T}} |\phi|^2}.$$

Proof intuition

For g_2^T , remove the whole turns made by a geodesic. What remains is the least mass of ω in a sliding arc of a fixed length r . Its average over the starting point is rL , so it cannot have minimum above rL . Fine periodic checkerboards make every sliding arc asymptotically have that mass. Equality means that every sliding arc has exactly its average mass; taking a distributional derivative says precisely that ω is invariant under rotation by r .

At times $2\pi m$, all distinct temporal Fourier modes of the circle are orthogonal. The full wave Gramian becomes block diagonal by Laplace eigenspaces, so its bottom eigenvalue is exactly $(T/2)g_1$. The quantity g_1 is controlled by the even Fourier coefficients of χ_ω . Fourier-flat unions of very small cells show that all these coefficients can be made uniformly small. Vanishing of every even coefficient, however, forces $\chi_\omega(x) + \chi_\omega(x + \pi)$ to be constant, which a $\{0, 1, 2\}$ -valued function permits only when $L = 1/2$.

The geometric functional at every time

Theorem 1. *Let $T > 0$ and $0 < L < 1$. Then*

$$\sup_{\omega \in \mathcal{U}_L} g_2^T(\omega) = L.$$

Write $T = 2\pi q + r$ with $q \in \mathbb{N} \cup \{0\}$ and $0 \leq r < 2\pi$. If $r = 0$, every $\omega \in \mathcal{U}_L$ is a maximizer. If $r > 0$, then ω is a maximizer if and only if

$$\chi_\omega(x+r) = \chi_\omega(x) \quad \text{for almost every } x. \quad (2)$$

Consequently maximizers exist for every L when $r/(2\pi)$ is rational, and no maximizer exists when $r/(2\pi)$ is irrational. Moreover, for every $\omega \in \mathcal{U}_L$,

$$g_2(\omega) := \lim_{T \rightarrow \infty} g_2^T(\omega) = L.$$

Proof. Put

$$F_r(x) = \int_0^r \chi_\omega(x+t) dt.$$

Every complete turn contributes $|\omega| = 2\pi L$, and therefore

$$g_2^T(\omega) = \frac{2\pi qL + \inf_x F_r(x)}{T}. \quad (3)$$

Fubini's theorem gives

$$\frac{1}{2\pi} \int_{\mathbb{T}} F_r(x) dx = rL.$$

Thus $\inf_x F_r(x) \leq rL$ (with essential infimum if preferred), proving $g_2^T(\omega) \leq L$.

For sharpness, divide the circle into N cells of length $2\pi/N$ and retain the first proportion L of each cell:

$$\omega_N = \bigcup_{k=0}^{N-1} \left[\frac{2\pi k}{N}, \frac{2\pi(k+L)}{N} \right).$$

Every arc differs from a union of complete cells only at its two ends. Hence, uniformly in x and $0 \leq r \leq 2\pi$,

$$|F_r^{\omega_N}(x) - rL| \leq \frac{4\pi}{N}.$$

Together with (3), this proves $g_2^T(\omega_N) \rightarrow L$.

Equality in the minimum–average bound holds exactly when F_r is constant almost everywhere. Its distributional derivative is

$$F'_r(x) = \chi_\omega(x+r) - \chi_\omega(x),$$

so equality is equivalent to (2). If $r/(2\pi) = p/s$ in lowest terms, the set

$$\bigcup_{k=0}^{s-1} \left[\frac{2\pi k}{s}, \frac{2\pi(k+L)}{s} \right)$$

is invariant under rotation by r and has measure $2\pi L$. If the ratio is irrational, ergodicity of irrational circle rotation forces every invariant measurable set to have measure zero or 2π , impossible here. Finally, (3) directly gives $g_2^T(\omega) \rightarrow L$ as $T \rightarrow \infty$. $\square$

The deterministic constant at resonant times

Lemma 2. *For every $0 < L < 1$, there are sets $\omega_N \in \mathcal{U}_L$ such that*

$$\sup_{\ell \in \mathbb{Z} \setminus \{0\}} \left| \int_{\omega_N} e^{i\ell x} dx \right| \rightarrow 0. \quad (4)$$

Proof. Partition the circle into N equal cells I_k . Select cells independently with probability L . The elementary Hoeffding bound, followed by a union bound over the $N - 1$ nonzero discrete Fourier frequencies, shows that one can choose a realization for which both the discrepancy in the number of selected cells and every discrete Fourier sum

$$\sum_{k \text{ selected}} e^{2\pi i a k / N}, \quad 1 \leq a < N,$$

are $O(\sqrt{N \log N})$. Add or remove $O(\sqrt{N \log N})$ cells to leave exactly $\lfloor LN \rfloor$ full cells, and add a subinterval of one remaining cell to make the measure exactly $2\pi L$. The same bounds persist.

For an integer $\ell \neq 0$, the integral over the union of full cells factors as

$$\left(\int_0^{2\pi/N} e^{i\ell x} dx \right) \sum_{k \text{ selected}} e^{2\pi i \ell k / N}.$$

The first factor has modulus at most $2\pi/N$; when N divides ℓ it is zero. The final partial cell contributes at most $2\pi/N$. Uniformly in $\ell \neq 0$, the result is $O(\sqrt{\log N / N})$, proving (4). $\square$

Theorem 3. *Let $m \in \mathbb{N}^*$, $T = 2\pi m$, and $0 < L < 1$. For every $\omega \in \mathcal{U}_L$,*

$$C_T(\omega) = \frac{T}{2} g_1(\omega), \quad (5)$$

where

$$g_1(\omega) = L - \frac{1}{2\pi} \sup_{j \geq 1} \left| \int_{\omega} e^{2ijx} dx \right|. \quad (6)$$

Consequently

$$\sup_{\omega \in \mathcal{U}_L} C_T(\omega) = \frac{TL}{2}.$$

This supremum is attained if and only if $L = 1/2$. For $L = 1/2$, the maximizers are exactly the antipodal selectors

$$\chi_{\omega}(x) + \chi_{\omega}(x + \pi) = 1 \quad a.e. \quad (7)$$

Proof. The positive circle frequencies are the integers $j \geq 1$, with eigenspaces

$$E_j = \text{span}\{\cos(jx), \sin(jx)\}.$$

In the source's y^+, y^- variables, integration over $[0, 2\pi m]$ kills every temporal term with nonzero integer frequency. The observation Gramian is therefore block diagonal: on each of the two time branches and each E_j , its quadratic form is $T \int_{\omega} |u|^2$, while the source normalization contributes the factor $1/2$. Taking the lowest block eigenvalue proves (5).

Every unit vector in E_j is a phase shift of $\pi^{-1/2} \sin(jx)$. Since

$$\sin^2(jx + \theta) = \frac{1}{2} - \frac{1}{2} \cos(2jx + 2\theta),$$

minimization over θ gives (6). Lemma 2 then yields $\sup g_1 = L$ and hence the stated supremum of C_T .

If $g_1(\omega) = L$, every even nonzero Fourier coefficient of χ_{ω} vanishes. On $[0, \pi)$ define

$$h(x) = \chi_{\omega}(x) + \chi_{\omega}(x + \pi).$$

All nonconstant Fourier coefficients of h vanish, so Fourier uniqueness gives $h = 2L$ almost everywhere. But h takes values only in $\{0, 1, 2\}$; as $0 < L < 1$, this forces $L = 1/2$ and $h = 1$. Conversely, (7) makes every even nonzero Fourier coefficient vanish and therefore gives $g_1 = L$. This proves the classification. $\square$

Upgrade attempts, limitations, and novelty bounds

The finite-time g_2^T theorem is a complete result for the circle and strictly strengthens its long-time g_2 subcase. The deterministic theorem settles an infinite family of observation times but not arbitrary T .

An additional upgrade attempt used rotation averaging: concavity and translation invariance give the natural relaxed upper bound obtained from the constant density L . The checkerboards in Theorem 1 also make the high-frequency geometric obstruction converge uniformly to this relaxed value. What remains for arbitrary T is a two-scale estimate controlling the off-diagonal wave Gramian when the observation cells and the Fourier modes both tend to high frequency. Weak-* convergence of characteristic functions does not control this operator norm, so the resonant proof cannot simply be perturbed. Higher-dimensional manifolds likewise lose the one-dimensional sliding-arc reduction.

On 11 August 2026, searches covered the exact source title and question, g_2^T /circle/sliding-window formulations, resonant-time deterministic observability, and the citing literature. Privat–Trélat–Zuazua [2] already note Parseval equality of deterministic and randomized constants at common periods in a one-dimensional interval and establish related no-gap/nonattainment results for the randomized spectral

design. No source located in the bounded search states Theorem 1 or this combined circle classification. Novelty confidence is moderate: the proofs are elementary once the circle reduction is made, and the resonant part has clear antecedents.

The included script numerically checks sliding-window convergence, Fourier-flat finite-cell samples, and the formulas for g_1 ; it is not used in the proof. Human review should focus on the normalization in (5), the uniform-in-frequency step in Lemma 2, and the distinction between essential and pointwise infima for arbitrary measurable representatives.

References

- [1] E. Humbert, Y. Privat, and E. Trélat, *Observability properties of the homogeneous wave equation on a closed manifold*, arXiv:1607.01535, concluding shape-optimization problems (PDF p. 22).
- [2] Y. Privat, E. Trélat, and E. Zuazua, *Optimal observability of the multi-dimensional wave and Schrödinger equations in quantum ergodic domains*, arXiv:1211.5655.
- [3] W. Hoeffding, *Probability inequalities for sums of bounded random variables*, J. Amer. Statist. Assoc. 58 (1963), 13–30.